

**MARK SCHEME for the May/June 2011 question paper**  
**for the guidance of teachers**

**9709 MATHEMATICS**

**9709/31**

Paper 3, maximum raw mark 75

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- Cambridge will not enter into discussions or correspondence in connection with these mark schemes.

Cambridge is publishing the mark schemes for the May/June 2011 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.



|               |                                       |                 |              |
|---------------|---------------------------------------|-----------------|--------------|
| <b>Page 2</b> | <b>Mark Scheme: Teachers' version</b> | <b>Syllabus</b> | <b>Paper</b> |
|               | <b>GCE AS/A LEVEL – May/June 2011</b> | <b>9709</b>     | <b>31</b>    |

## **Mark Scheme Notes**

Marks are of the following three types:

**M** Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

**A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

**B** Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more “method” steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep\*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- The symbol  $\surd$  implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously “correct” answers or results obtained from incorrect working.
- Note: B2 or A2 means that the candidate can earn 2 or 0.  
B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.
- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking  $g$  equal to 9.8 or 9.81 instead of 10.

|               |                                       |                 |              |
|---------------|---------------------------------------|-----------------|--------------|
| <b>Page 3</b> | <b>Mark Scheme: Teachers' version</b> | <b>Syllabus</b> | <b>Paper</b> |
|               | <b>GCE AS/A LEVEL – May/June 2011</b> | <b>9709</b>     | <b>31</b>    |

The following abbreviations may be used in a mark scheme or used on the scripts:

|     |                                                                                                                                                                                         |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AEF | Any Equivalent Form (of answer is equally acceptable)                                                                                                                                   |
| AG  | Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)                                                     |
| BOD | Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)                                                                                                  |
| CAO | Correct Answer Only (emphasising that no “follow through” from a previous error is allowed)                                                                                             |
| CWO | Correct Working Only – often written by a ‘fortuitous’ answer                                                                                                                           |
| ISW | Ignore Subsequent Working                                                                                                                                                               |
| MR  | Misread                                                                                                                                                                                 |
| PA  | Premature Approximation (resulting in basically correct work that is insufficiently accurate)                                                                                           |
| SOS | See Other Solution (the candidate makes a better attempt at the same question)                                                                                                          |
| SR  | Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance) |

### **Penalties**

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MR –1 | A penalty of MR –1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become “follow through ✓” marks. MR is not applied when the candidate misreads his own figures – this is regarded as an error in accuracy. An MR –2 penalty may be applied in particular cases if agreed at the coordination meeting. |
| PA –1 | This is deducted from A or B marks in the case of premature approximation. The PA –1 penalty is usually discussed at the meeting.                                                                                                                                                                                                                                                                                                                             |

| Page 4 | Mark Scheme: Teachers' version | Syllabus | Paper |
|--------|--------------------------------|----------|-------|
|        | GCE AS/A LEVEL – May/June 2011 | 9709     | 31    |

- 1 Either: Obtain  $1 + \frac{1}{3}kx$ , where  $k = \pm 6$  or  $\pm 1$  M1  
Obtain  $1 - 2x$  A1  
Obtain  $-4x^2$  A1  
Obtain  $-\frac{40}{3}x^3$  or equivalent A1
- Or: Differentiate expression to obtain form  $k(1 - 6x)^{-\frac{2}{3}}$  and evaluate  $f(0)$  and  $f'(0)$  M1  
Obtain  $f'(x) = -2(1 - 6x)^{-\frac{2}{3}}$  and hence the correct first two terms  $1 - 2x$  A1  
Obtain  $f''(x) = -8(1 - 6x)^{-\frac{5}{3}}$  and hence  $-4x^2$  A1  
Obtain  $f'''(x) = -80(1 - 6x)^{-\frac{8}{3}}$  and hence  $-\frac{40}{3}x^3$  or equivalent A1 [4]
- 2 (i) Obtain  $\frac{k \cos 2x}{1 + \sin 2x}$  for any non-zero constant  $k$  M1  
Obtain  $\frac{2 \cos 2x}{1 + \sin 2x}$  A1 [2]
- (ii) Use correct quotient or product rule M1  
Obtain  $\frac{x \sec^2 x - \tan x}{x^2}$  or equivalent A1 [2]
- 3 (i) Obtain  $\pm \begin{pmatrix} 3 \\ -4 \\ 6 \end{pmatrix}$  as normal to plane B1  
Form equation of  $p$  as  $3x - 4y + 6z = k$  or  $-3x + 4y - 6z = k$  and use relevant point to find  $k$  M1  
Obtain  $3x - 4y + 6z = 80$  or  $-3x + 4y - 6z = -80$  A1 [3]
- (ii) State the direction vector  $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$  or equivalent B1  
Carry out correct process for finding scalar product of two relevant vectors M1  
Use correct complete process with moduli and scalar product and evaluate  $\sin^{-1}$  or  $\cos^{-1}$  of result M1  
Obtain  $30.8^\circ$  or  $0.538$  radians A1 [4]

| Page 5 | Mark Scheme: Teachers' version | Syllabus | Paper |
|--------|--------------------------------|----------|-------|
|        | GCE AS/A LEVEL – May/June 2011 | 9709     | 31    |

- 4 (i) Verify that  $-96 + 100 + 8 - 12 = 0$  B1  
 Attempt to find quadratic factor by division by  $(x + 2)$ , reaching a partial quotient  
 $12x^2 + kx$ , inspection or use of an identity M1  
 Obtain  $12x^2 + x - 6$  A1  
 State  $(x + 2)(4x + 3)(3x - 2)$  A1 [4]  
 [The M1 can be earned if inspection has unknown factor  $Ax^2 + Bx - 6$  and an equation in  $A$  and/or  $B$  or equation  $12x^2 + Bx + C$  and an equation in  $B$  and/or  $C$ .]
- (ii) State  $3^y = \frac{2}{3}$  and no other value B1  
 Use correct method for finding  $y$  from equation of form  $3^y = k$ , where  $k > 0$  M1  
 Obtain  $-0.369$  and no other value A1 [3]
- 5 (i) Use at least one of  $e^{2x} = 9$ ,  $e^y = 2$  and  $e^{2y} = 4$  B1  
 Obtain given result  $58 + 2k = c$  AG B1 [2]
- (ii) Differentiate left-hand side term by term, reaching  $ae^{2x} + be^y \frac{dy}{dx} + ce^{2y} \frac{dy}{dx}$  M1  
 Obtain  $12e^{2x} + ke^y \frac{dy}{dx} + 2e^{2y} \frac{dy}{dx}$  A1  
 Substitute  $(\ln 3, \ln 2)$  in an attempt involving implicit differentiation at least once, where  
 RHS = 0 M1  
 Obtain  $108 - 12k - 48 = 0$  or equivalent A1  
 Obtain  $k = 5$  and  $c = 68$  A1 [5]
- 6 (i) State or imply area of segment is  $\frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta$  or  $50\theta - 50\sin\theta$  B1  
 Attempt to form equation from area of segment =  $\frac{1}{5}$  of area of circle, or equivalent M1  
 Confirm given result  $\theta = \frac{2}{5}\pi + \sin\theta$  A1 [3]
- (ii) Use iterative formula correctly at least once M1  
 Obtain value for  $\theta$  of 2.11 A1  
 Show sufficient iterations to justify value of  $\theta$  or show sign change in interval  
 (2.105, 2.115) A1  
 Use correct trigonometry to find an expression for the length of  $AB$  M1  
 e.g.  $20 \sin 1.055$  or  $\sqrt{200 - 200 \cos 2.11}$   
 Hence 17.4 A1 [5]  
 [2.1  $\rightarrow$  2.1198  $\rightarrow$  2.1097  $\rightarrow$  2.1149  $\rightarrow$  2.1122]

| Page 6 | Mark Scheme: Teachers' version | Syllabus | Paper |
|--------|--------------------------------|----------|-------|
|        | GCE AS/A LEVEL – May/June 2011 | 9709     | 31    |

- 7 (i) State or imply  $dx = 2t dt$  or equivalent B1  
Express the integral in terms of  $x$  and  $dx$  M1  
Obtain given answer  $\int_1^5 (2x - 2) \ln x dx$ , including change of limits AG A1 [3]
- (ii) Attempt integration by parts obtaining  $(ax^2 + bx) \ln x \pm \int (ax^2 + bx) \frac{1}{x} dx$  or equivalent M1  
Obtain  $(x^2 - 2x) \ln x - \int (x^2 - 2x) \frac{1}{x} dx$  or equivalent A1  
Obtain  $(x^2 - 2x) \ln x - \frac{1}{2} x^2 + 2x$  A1  
Use limits correctly having integrated twice M1  
Obtain  $15 \ln 5 - 4$  or exact equivalent A1 [5]  
[Equivalent for M1 is  $(2x - 2)(ax \ln x + bx) - \int (ax \ln x + bx) 2dx$ ]
- 8 (i) Either: Multiply numerator and denominator by  $(1 - 2i)$ , or equivalent M1  
Obtain  $-3i$  A1  
State modulus is 3 A1  
Refer to  $u$  being on negative imaginary axis or equivalent and confirm argument as  $-\frac{1}{2}\pi$  A1
- Or: Using correct processes, divide moduli of numerator and denominator M1  
Obtain 3 A1  
Subtract argument of denominator from argument of numerator M1  
Obtain  $-\tan^{-1} \frac{1}{2} - \tan^{-1} 2$  or  $-0.464 - 1.107$  and hence  $-\frac{1}{2}\pi$  or  $-1.57$  A1 [4]
- (ii) Show correct half-line from  $u$  at angle  $\frac{1}{4}\pi$  to real direction B1  
Use correct trigonometry to find required value M1  
Obtain  $\frac{3}{2}\sqrt{2}$  or equivalent A1 [3]
- (iii) Show, or imply, locus is a circle with centre  $(1 + i)u$  and radius 1 M1  
Use correct method to find distance from origin to furthest point of circle M1  
Obtain  $3\sqrt{2} + 1$  or equivalent A1 [3]

| Page 7 | Mark Scheme: Teachers' version | Syllabus | Paper |
|--------|--------------------------------|----------|-------|
|        | GCE AS/A LEVEL – May/June 2011 | 9709     | 31    |

- 9 (i) Express  $\cos 4\theta$  as  $2 \cos^2 2\theta - 1$  or  $\cos^2 2\theta - \sin^2 2\theta$  or  $1 - 2 \sin^2 2\theta$  B1  
Express  $\cos 4\theta$  in terms of  $\cos \theta$  M1  
Obtain  $8 \cos^4 \theta - 8 \cos^2 \theta + 1$  A1  
Use  $\cos 2\theta = 2 \cos^2 \theta - 1$  to obtain given answer  $8 \cos^4 \theta - 3$  AG A1 [4]
- (ii) (a) State or imply  $\cos^4 \theta = \frac{1}{2}$  B1  
Obtain 0.572 B1  
Obtain  $-0.572$  B1 [3]
- (b) Integrate and obtain form  $k_1 \theta + k_2 \sin 4\theta + k_3 \sin 2\theta$  M1  
Obtain  $\frac{3}{8} \theta + \frac{1}{32} \sin 4\theta + \frac{1}{4} \sin 2\theta$  A1  
Obtain  $\frac{3}{32} \pi + \frac{1}{4}$  following completely correct work A1 [3]
- 10 (i) Separate variables correctly and integrate of at least one side M1  
Carry out an attempt to find  $A$  and  $B$  such that  $\frac{1}{N(1800 - N)} \equiv \frac{A}{N} + \frac{B}{1800 - N}$ , or equivalent M1  
Obtain  $\frac{2}{N} + \frac{2}{1800 - N}$  or equivalent A1  
Integrates to produce two terms involving natural logarithms M1  
Obtain  $2 \ln N - 2 \ln (1800 - N) = t$  or equivalent A1  
Evaluate a constant, or use  $N = 300$  and  $t = 0$  in a solution involving  $a \ln N$ ,  $b \ln(1800)$  and  $ct$  M1  
Obtain  $2 \ln N - 2 \ln (1800 - N) = t - 2 \ln 5$  or equivalent A1  
Use laws of logarithms to remove logarithms M1  
Obtain  $N = \frac{1800e^{\frac{1}{2}t}}{5 + e^{\frac{1}{2}t}}$  or equivalent A1 [9]
- (ii) State or imply that  $N$  approaches 1800 B1 [1]